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Roll No

MI-5003-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Mineral Processing

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any Five questions.
ii) All questions carry equal marks.

1. a) What do you understand by Pelletisation? Describe with neat sketch any one pelletisation technique.
b) Draw a flow sheet of crushing and grinding plant.
2. a) Discuss in detail the theory of ball mill operation.
b) Discuss the scope, objectives and economics of mineral processing also state its advantages.
3. Describe different types of screen used in mineral industry, their application and limitations.
4. Write notes on the following:
 - a) Sorting classifier
 - b) Spiral concentrator
5. a) Name the different classes of classifiers. What is Hydro Cyclone? Name the different parts of hydro cyclones neat sketch, explain their functions.
b) With the help of a flow sheet describe the preliminary treatment of the sulphide ore of copper.

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6.
 - a) Name the different types of magnetic separators explain with neat sketch dry belt magnetic separators.
 - b) Explain working principle and constructional features of a shaking table.

7.
 - a) Define the terms :
 - i) Enrichment ratio
 - ii) Percentage recovery
 - iii) Ratio of concentration
 - iv) Efficiency of screen
 - b) Describe the working principle of a Baum Jig.

8. Write short notes on any four of the following:
 - a) Electrostatic separator
 - b) Thickener
 - c) Modifiers in froth floatation
 - d) Sauter mean diameter of particles
 - e) Sampling on a heap of fines ore
 - f) Tube mills.
